Performance Medicine: *a scoping review of integrated human performance systems & the need for conceptual architecture*

University of St Andrews

strikerselite
sports injury group
dublin

Lewis Turmeau, Paul Dijkstra, Ronan Kearney, Andrew Murray, Pieter D'Hooghe, Eva Carneiro, Stuart O'Flanagan, Fearghal Kerin, Edel Fanning, Rajeev Routray, Carrie McCrea, Claire-Marie Roberts, Calum Arthur, Alan Kourie, Jack Wilby, Andrea Goceva, Ciara Grant, Andrew J Hall

Introduction

Performance Medicine describes the integration of health optimisation and performance enhancement across high-demand human systems.

Related models are used in elite sport and tactical, military, and corporate settings, yet the field lacks an agreed definition, conceptual structure, and governing framework. Without these foundations the design, delivery, standardisation, evaluation, and development of PM is limited.

Objectives:

- Map the literature relating to integrated health & performance systems
- Examine how these systems are described & delivered
- Determine whether a coherent conceptual structure for PM exists

Methods

Design: PRISMA-ScR scoping review of the literature
Sources: PubMed, Scopus, ProQuest Central, DOAJ, Google Scholar
Scope: 2000-2025, English language
Types: Empirical studies, reviews, frameworks, models
Setting: Integrated health systems in high-performance groups
Criteria: Prospectively-applied inclusion & exclusion criteria
Variables: Frequency, case mix, context associations, service design
Analysis: Descriptive, thematic analysis, stakeholder synthesis.

A structured claim-evidence synthesis was applied across the included literature, with each claim anchored to identified sources & weighted by evidence type and strength

Results: Literature Characteristics

Most sources were descriptive or practice-led rather than theory-building, reflecting an active but disparate field without clearly conceptual architecture.

- Mostly low-level evidence (expert opinion, case descriptions)
- Heterogeneous terminology & study methods, limiting comparability
- Content & focus varied across disciplines & settings
- No unifying framework, taxonomy, or governance structure

106 Included Studies

Descriptive sources predominate

Multidisciplinary academic & applied field

Inconsistent terminology

Limited methodological comparability

Widespread practice is evident

Growing interest in field development

Results: What is Performance Medicine?

Integrated systems for health and performance optimisation is widespread in elite sport, common military and tactical settings, and emerging in corporate and technical contexts e.g. healthcare, engineering, and technology sectors.

The literature suggests PM requires:

- Integration of medical, technical & supporting disciplines
- Healthcare for general health + performance
- System coordination is important for success
- Highly goal-oriented & context-specific

Overlap, but distinctions, with related holistic fields (Table1):

- Performance Medicine
- Sport & Exercise Medicine
- Lifestyle Medicine

Results: Performance Medicine & related fields

Dimension	Performance Medicine	Sports Medicine	Lifestyle Medicine	Overlap with PM
Primary focus	Integration of health optimisation and performance enhancement across high-demand human systems	Diagnosis, treatment, and prevention of sports-related injury and illness	Prevention and reversal of chronic lifestyle-related disease through behavioural change	*All share a clinical and preventative orientation*
Core disciplines	Medicine · Psychology · Allied health · Performance science (interdisciplinary, integrated)	Sports medicine · Physiotherapy · Imaging · Surgery	Medicine · Nutrition · Exercise · Psychology · Behaviour change	*SEM and LM both contribute domain expertise to PM*
Population	Athletes, military personnel, corporate/technical performers — high-demand human systems	Athletes and physically active populations	General population; those with or at risk of chronic disease	*PM includes athletic populations (overlap with SEM)*
Primary goal	Optimise readiness, resilience, and sustained performance while maintaining health	Return to sport; preserve musculoskeletal health and athletic function	Prevent and treat chronic disease; improve quality of life	*Health as the substrate for performance (PM) — shared value*
System model	Layered architecture: domains → components → governance → context (collective, integrated)	Largely episodic and consultation-based; team models emerging	Preventative pathway model; patient-centred; behaviour-change focus	*PM is defined by its integration of multiple models*
Governance / accreditation	No unified framework or accreditation yet identified — primary evidence gap (C6)	FSEM · CSEM · National sports medicine bodies · WADA	ACLM · BSLM · National equivalents · Primary care pathways	*PM draws on SEM and LM professional standards*
Evidence base	Predominantly conceptual and low-level (Level III–V); no randomised trials for the integrated model	Robust within-domain; systematic reviews and RCTs for many conditions	Growing base; RCTs for lifestyle interventions; primary care evidence	*SEM and LM provide stronger within-domain evidence than PM overall*
Conceptual distinctiveness	Distinct by virtue of integrated architecture, performance orientation, and system-level governance (C7)	Medically-led; primarily injury and illness focus; less performance-system emphasis	Disease-prevention and behaviour-change focus; not performance-oriented	*Overlap exists but does not render PM indistinct (C7)*

Conclusion

Performance Medicine is widely practised but insufficiently defined.
Lack of conceptual architecture limits standardisation, evaluation & best practice.
Empirical study and multidisciplinary stakeholder synthesis is needed to:

- Define the field more clearly
- Develop an integrated framework
- Build operational tools for PM system design, delivery & governance.

Scan to explore & contribute

Scan the to explore our interactive research page, and help shape the field with the ***Performance Medicine Stakeholder Survey*** (5 to 8 minutes).

Your perspective matters.

SCAN TO CONTRIBUTE